

Lab03_Padovani

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Laboratório 03

Dados relacionais e operações do tipo join

```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.1    ✓ readr      2.2.0
✓ forcats    1.0.1    ✓ stringr    1.6.0
✓ ggplot2    4.0.3    ✓ tibble     3.3.1
✓ lubridate  1.9.5    ✓ tidyr      1.3.2
✓ purrr      1.2.2

— Conflicts — tidyverse_conflicts() —
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(leaflet)
```

1 - Importe, utilizando o pacote readr, cada um dos três arquivos disponíveis. Os objetos resultantes devem ser chamados flights, airlines e airports.

```
airlines <- read_csv("Data/airlines.csv")
```

Rows: 14 Columns: 2

— Column specification —

Delimiter: ","

chr (2): IATA_CODE, AIRLINE

ℹ Use `spec()` to retrieve the full column specification for this data.

ℹ Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
airports <- read_csv("Data/airports.csv",
                     col_select = c(IATA_CODE, CITY, STATE, LATITUDE, LONGITUDE))
```

Rows: 322 Columns: 5

— Column specification —

Delimiter: ","

chr (3): IATA_CODE, CITY, STATE

dbl (2): LATITUDE, LONGITUDE

ℹ Use `spec()` to retrieve the full column specification for this data.

ℹ Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
processa_lote_voos <- function(df, pos) {
  df %>%
    filter(!str_starts(DESTINATION_AIRPORT, "1")) %>%
    drop_na() %>%
    group_by(DESTINATION_AIRPORT) %>%
    summarise(
      soma_atrasos = sum(ARRIVAL_DELAY),
      qtd_voos = n(),
      .groups = "drop"
    )
}
```

```
cols <- cols_only(
  DESTINATION_AIRPORT = col_character(),
  ARRIVAL_DELAY = col_double()
)

flights <- read_csv_chunked(
  file = "Data/flights.csv",
  callback = DataFrameCallback$new(processa_lote_voos),
  chunk_size = 1000000,
  col_types = cols
)
```

```
flights_summary <- flights %>%
  group_by(DESTINATION_AIRPORT) %>%
  summarise(
    total_atrasos = sum(soma_atrasos),
    total_voos = sum(qtd_voos),
    .groups = "drop"
  ) %>%
  mutate(ATRASO_MEDIO = total_atrasos / total_voos) %>%
  select(DESTINATION_AIRPORT, ATRASO_MEDIO)
```

```
flights_summary
```

```
# A tibble: 322 × 2
```

	DESTINATION_AIRPORT	ATRASO_MEDIO
	<chr>	<dbl>
1	ABE	5.80
2	ABI	4.08
3	ABQ	5.61
4	ABR	-3.39
5	ABY	8.68
6	ACK	4.91
7	ACT	10.6
8	ACV	9.68
9	ACY	11.8
10	ADK	-3.19

```
# i 312 more rows
```

2 - Selecione a operação apropriada join para incluir, na tabela flights, as colunas CITY e STATE do objeto airports. Para executar esta tarefa:

```
aeroportos_ausentes <- anti_join(  
  flights_summary,  
  airports,  
  by = c("DESTINATION_AIRPORT" = "IATA_CODE")  
)  
print(aeroportos_ausentes)
```

A tibble: 0 × 2

i 2 variables: DESTINATION_AIRPORT <chr>, ATRASO_MEDIO <dbl>

```
flights <- left_join(  
  flights_summary,  
  airports,  
  by = c("DESTINATION_AIRPORT" = "IATA_CODE")  
)  
flights
```

A tibble: 322 × 6

	DESTINATION_AIRPORT	ATRASO_MEDIO	CITY	STATE	LATITUDE	LONGITUDE
	<chr>	<dbl>	<chr>	<chr>	<dbl>	<dbl>
1	ABE	5.80	Allentown	PA	40.7	-75.4
2	ABI	4.08	Abilene	TX	32.4	-99.7
3	ABQ	5.61	Albuquerque	NM	35.0	-107.
4	ABR	-3.39	Aberdeen	SD	45.4	-98.4
5	ABY	8.68	Albany	GA	31.5	-84.2
6	ACK	4.91	Nantucket	MA	41.3	-70.1
7	ACT	10.6	Waco	TX	31.6	-97.2
8	ACV	9.68	Arcata/Eureka	CA	41.0	-124.
9	ACY	11.8	Atlantic City	NJ	39.5	-74.6
10	ADK	-3.19	Adak	AK	51.9	-177.

i 312 more rows

3 - Quantos aeroportos cada estado possui? Apresente uma tabela ordenada de forma decrescente (no número de aeroportos).

```
tabela_estados <- airports %>%  
  group_by(STATE) %>%  
  summarise(numero_aeroportos = n()) %>%  
  arrange(desc(numero_aeroportos))  
  
print("Número de aeroportos por Estado:")
```

[1] "Número de aeroportos por Estado:"

```
print(tabela_estados)
```

A tibble: 54 × 2

STATE numero_aeroportos

	<chr>	<int>
1	TX	24
2	CA	22
3	AK	19
4	FL	17
5	MI	15
6	NY	14
7	CO	10
8	MN	8
9	MT	8
10	NC	8

i 44 more rows

4 - Apresente um mapa representando todos os atrasos observados por aeroporto.

```
flights_mapa <- flights %>% drop_na(LATITUDE, LONGITUDE)

# Criando o mapa com leaflet
this <- leaflet(data = flights_mapa) %>%
  addTiles() %>%
  addMarkers(
    lng = ~LONGITUDE,
    lat = ~LATITUDE,
    popup = ~paste0(
      "<b>Aeroporto:</b> ", DESTINATION_AIRPORT, " - ", CITY, "<br>",
      "<b>Atraso Médio:</b> ", round(ATRASO_MEDIO, 2), " min"
    )
  )
```

this



```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-08-31 23:03:57 -03"
```